

Does Transfer Spending Buy Success in La Liga?

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Motivation

I chose this topic because transfer spending is one of the biggest debates in football. Every summer, fans argue that some clubs are basically "buying success," but the actual relationship is not always that simple. Some clubs consistently overperform their budgets, while others spend heavily and still finish disappointingly. I wanted to actually check whether the link between money spent and success on the pitch is as strong as people assume, or whether on-pitch performance metrics tell a different story. La Liga was a good choice for this because it has a long history of two dominant spenders (Real Madrid and Barcelona) along with a wide spread of mid-table clubs operating on very different budgets.

Data Source

The project uses two main data sources.

The first is a transfer dataset from the public GitHub repository [d2ski/football-transfers-data](https://github.com/d2ski/football-transfers-data), which contains transfer records scraped from Transfermarkt for La Liga clubs from 2009 to 2021. Each row represents one transfer and includes the club, player, transfer fee in euros, direction (incoming or outgoing), and whether it was a loan or a paid transfer. I filtered for permanent transfers with actual fees only, since loans and free transfers don't really represent cash movement.

At first I wanted to use FBref for the performance data because it had richer statistics, especially xG and xGA, but I could not get around the scraping restrictions in a clean way. Because of that, I switched to football-data.co.uk, which gives La Liga match results as clean CSV files for every season. From the match results I computed final standings manually (points, wins, draws, losses, goals for, goals against) along with additional features like goal difference, points per game, and win rate.

One annoying part was standardizing club names, because the same club appeared with different names across the two datasets — for example "Athletic Club" in one and "Athletic Bilbao" in the other. I had to write a normalization function and a manual mapping to fix this before merging.

After merging on club name and season, the final dataset contains 140 club-season observations spanning 2014-2015 through 2020-2021.

Data Analysis

I started with exploratory data analysis. Net transfer spend turned out to be heavily right-skewed, with Real Madrid and Barcelona accounting for a huge share of total spending and most other clubs sitting closer to zero. Points are roughly normally distributed around the mid-40s, which is what you'd expect over a 38-game season. The scatter plot of net spend vs final points showed a positive trend but with a lot of variance, which suggested the relationship is real but not deterministic. Tracking spending groups across seasons confirmed that the top 5 spenders usually finish with significantly more points than the bottom 5, but there are clear outliers in both directions.

For hypothesis testing I ran two tests. The Spearman correlation between net transfer spend and final points gave $\rho = 0.155$ with a p-value of 0.0675, so at the 5% level I could not reject the null hypothesis. The relationship is positive but weaker than I expected when looking at just spend and points directly. The Mann-Whitney U test, which compared top-half versus bottom-half spenders within each season, gave a p-value of 0.0019, so I rejected the null hypothesis at the 5% level. The two tests pointing in different directions is itself interesting — it suggests that the rank-based correlation across all clubs is noisy, but the difference between high spenders and low spenders as groups is real and significant.

For the machine learning part I trained a Random Forest and a Ridge Regression to predict final points. In the first model, the **Random Forest reached $R^2 = 0.957$** , but this was misleading because goal difference and win rate are directly connected to the final points — they are essentially measuring the same thing as the target. After feedback from the TA, I built a second model using only information available before the season: net transfer spend, previous season points, and previous season goal difference. In the pre-season model, the **Random Forest R^2 dropped to 0.508** and Ridge to **0.577**, with cross-validated R^2 of 0.517 and 0.547 respectively. Even though the score was worse, I think this version is more honest because it only uses information that would actually be known before the season starts.

Findings

The basic finding is that transfer spending and league success are related, but spending alone is not a strong predictor of where a club finishes. The Spearman correlation between net spend and points was positive but not significant at the 5% level ($\rho = 0.155$, $p = 0.0675$), while the group comparison via Mann-Whitney was clearly significant ($p = 0.0019$). Together, these results suggest that spending matters when you look at clubs in aggregate spending groups, but the relationship isn't strong enough to predict individual club outcomes from spending alone.

The more interesting finding came from the pre-season model. When restricted to features known before the season started, the previous season's points came out as the strongest predictor of next season's points, with net transfer spend as a secondary factor. In other words, knowing how a club performed last year tells you more than knowing how much they spent this summer. Money helps clubs maintain or slightly improve their position, but it rarely transforms a mid-table club into a title contender in a single season.

The outlier analysis brought up some interesting cases. For example, Atlético Madrid often finished higher than their transfer spending alone would predict, while Valencia appeared as an underperformer in several seasons. I did not model coaching or squad stability directly, but these cases suggest that factors like manager quality, club structure, and squad cohesion matter too.

Limitations and Future Work

The dataset size was one of the biggest weaknesses of the project. I originally expected to work with more seasons, but after merging the two sources, the overlap was only seven seasons. That limits the statistical power of the models, especially the pre-season one which loses even more rows because of the lagging. The borderline Spearman p-value (0.0675) is a good example — with more seasons of data, that result might have either become clearly significant or clearly not, instead of being right on the edge.

The analysis also doesn't include some features I would have liked to use. xG and xGA from FBref would have given a better picture of underlying performance, but the scraping issue made that data inaccessible during the project window. Squad market value (separate from transfer spend) would also be a more accurate measure of a club's actual financial position, since clubs with already-valuable squads don't need to spend as much to stay competitive.

If I extended this project, I would first try to find or build a newer transfer dataset, because the transfer dataset I used stops at 2021. I would also try to bring in manager tenure or tactical metrics to explain the outlier cases. Doing the same analysis on the Premier League, Bundesliga, or Serie A would also be interesting to see whether La Liga's two-club dominance makes the money-success relationship look different from other leagues.

AI Usage Disclosure

Claude was used for assistance with coding and minor editing. The research question, data choices, analysis decisions, and interpretation of results are my own.